



# NVIDIA DGX B300

Setting a new bar for real-time AI performance, from training to inference.



## Delivering Unprecedented Efficiency to Every Enterprise

A dual transformation is forging the future of business. First, we've entered the era of AI reasoning, where models perform complex, multi-step thinking. Simultaneously, the data center is evolving into an AI factory, a new class of facility built to manufacture intelligence at scale. This new reality requires a fundamental rethinking of enterprise infrastructure.

For enterprises, this dual shift presents profound challenges. AI reasoning requires immense computation, memory, and bandwidth. At the same time, building and operationalizing an AI factory is complex, and many companies struggle with critical gaps in expertise, system integration, and rising energy costs. They're finding they lack the specialized teams and tools needed to run this sophisticated new ecosystem with the efficiency and resilience of a hyperscaler.

NVIDIA DGX™ B300 is the foundational building block for this era of AI, empowering pioneers to build their own AI factories capable of tackling the demands of AI reasoning. Powered by the [NVIDIA Blackwell Ultra architecture](#), DGX B300 is an AI powerhouse purpose-built to deliver hyperscaler performance in an enterprise-sized footprint—144 petaFLOPS of FP4 inference performance for reasoning workloads. Its redesigned, [NVIDIA MGX™](#)-compliant, air-cooled chassis allows for seamless integration into modern data centers. Paired with NVIDIA Mission Control software, DGX B300 simplifies AI operations, delivering the full-stack solution enterprises need to master the complexity of [generative AI](#) and unlock the return on their investment.

## Real-Time AI Powerhouse

DGX B300 is engineered to be the foundational building block of the AI factory. It enables AI innovators of all sizes to manufacture intelligence at scale, harnessing generative AI capabilities previously reserved for global-scale AI organizations. As a fully integrated system powered by NVIDIA Blackwell Ultra GPUs, NVIDIA® ConnectX®-8 networking, and NVIDIA Mission Control software, DGX B300 delivers unprecedented performance and hyperscale-grade efficiency. By combining exceptional training performance with leadership-class, real-time inference, DGX B300 empowers every organization to build scalable infrastructure for the era of AI reasoning.

## Key Features

- > Built with NVIDIA Blackwell Ultra GPUs
- > 2.3 TB of GPU memory space
- > 72 petaFLOPS of training performance
- > 144 petaFLOPS of inference performance
- > NVIDIA networking
- > Intel Xeon 6776P processors
- > Foundation of [NVIDIA DGX BasePOD™](#) and [NVIDIA DGX SuperPOD™](#)
- > Leverages [NVIDIA AI Enterprise](#) and NVIDIA Mission Control™ software

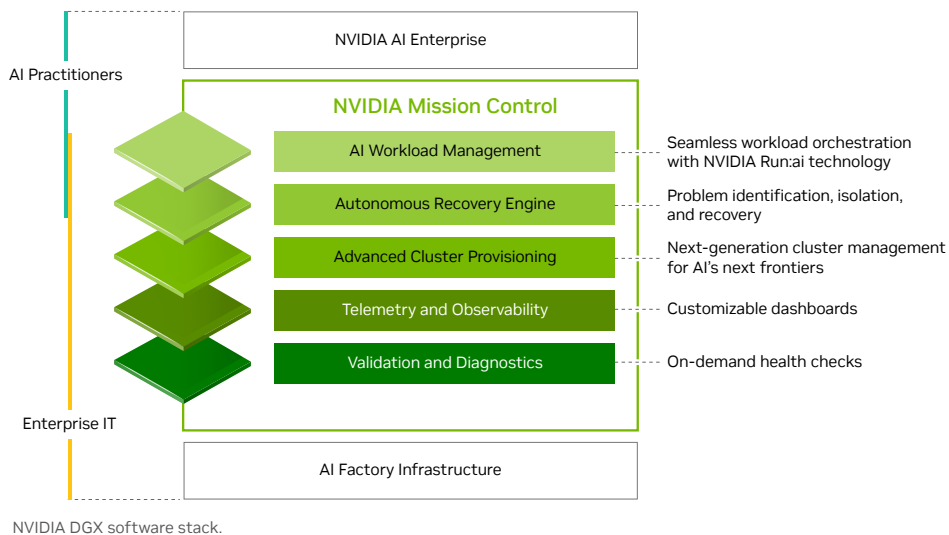
## A Blueprint for Modern Data Centers

DGX B300 serves as the blueprint for the modern AI factory, starting with its redesigned, Open Compute Project (OCP)-compliant chassis that brings hyperscaler design principles to any data center. For the first time, DGX B300 can be deployed using an NVIDIA MGX-compatible chassis or a more traditional AC-powered design, ensuring DGX B300 can be used in any infrastructure environment. By pairing this leading-edge design with practical serviceability, DGX B300 enables enterprises to construct AI factories on their own terms, with unprecedented efficiency and choice.

## Run Models, Automate the Essentials With NVIDIA Mission Control

Deploying an AI factory is more than an infrastructure purchase—it's an operational commitment. Many enterprises starting their AI transformation face significant complexity that leads to costly downtime and low utilization, directly hindering ROI. This is the challenge [NVIDIA Mission Control](#) is designed to solve.

Mission Control acts as a software-defined operations team, delivering the skills of a world-class AI factory operator in software to ensure enterprises get the most from their investment. It automates the full range of complex tasks—from initial cluster bring-up to daily workload management—allowing IT teams to run AI infrastructure with hyperscale-grade efficiency. To protect hardware investments, Mission Control provides critical infrastructure resiliency and maximizes productivity and uptime for AI factories, empowering developers to spend less time waiting and more time innovating.



DGX B300 Technical Specifications

|                              | DGX B300                                                                                                                                                                                                                                                 |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GPU                          | NVIDIA Blackwell Ultra GPUs                                                                                                                                                                                                                              |
| Total GPU Memory             | 2.3 TB                                                                                                                                                                                                                                                   |
| Performance                  | 144 PFLOPS FP4 inference*<br>72 PFLOPS FP8 training*                                                                                                                                                                                                     |
| NVIDIA NVLink™ Switch System | 2x                                                                                                                                                                                                                                                       |
| NVIDIA NVLink™ Bandwidth     | 14.4 TB/s aggregate bandwidth                                                                                                                                                                                                                            |
| Power Consumption            | ~14 kW                                                                                                                                                                                                                                                   |
| CPU                          | Intel Xeon 6776P processors                                                                                                                                                                                                                              |
| Networking                   | 8x OSFP ports serving 8x NVIDIA ConnectX-8 VPI<br>>Up to 800 Gb/s of NVIDIA InfiniBand/Ethernet<br>2x dual-port QSFP112 NVIDIA BlueField®-3 DPU<br>>Up to 400 Gb/s of NVIDIA InfiniBand/Ethernet                                                         |
| Management Network           | 1GbE onboard network interface card (NIC) with RJ45<br>1GbE RJ45 host baseboard management controller (BMC)                                                                                                                                              |
| Storage                      | OS: 2x 1.9 TB NVMe M.2<br>Internal storage: 8x 3.84 TB NVMe E1.S                                                                                                                                                                                         |
| Software                     | NVIDIA AI Enterprise (optimized AI software)<br><br>NVIDIA Mission Control (AI data center operations and orchestration with NVIDIA Run:ai technology)<br><br>NVIDIA DGX OS (operating system)<br><br>Supports Red Hat Enterprise Linux / Rocky / Ubuntu |
| Rack Units                   | 10                                                                                                                                                                                                                                                       |
| Operating Temperature        | 10C°–35C°                                                                                                                                                                                                                                                |
| Support                      | Three-year business-standard hardware and software support                                                                                                                                                                                               |

\*Shown with sparsity.

Ready to Get Started?

To learn more about DGX B300, visit [nvidia.com/dgx-b300](https://nvidia.com/dgx-b300)

